

IIT GUWAHATI
DEPARTMENT OF CSE
Quiz 01

CS207 DESIGN AND ANALYSIS OF ALGORITHMS

Time: 09:00am- 09:45am

FEB. 19

Max Marks: 30

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Invigilator [Signature]

MULTIPLE CHOICE QUESTIONS

Question 1 Which of the following is NOT a step in Strassen's matrix multiplication algorithm?

- ☐ A Divide matrices into submatrices
- ☒ B Transpose the matrices
- ☐ C Combine products to get the final result
- ☒ D Calculate seven products recursively

Question 2 In GS algorithm where women propose to men and preference lists start from index 0, consider a woman w who is matched with a man m who is her 5th choice index on her preference list. How many men have rejected w before she was matched with m ?

- ☐ A 5 ☐ B 0 ☐ C 6 ☐ D 4

Question 3 S1: There can be more than one possible stable matchings for a given preference list of men and women S2: The Gale-Shapley algorithm to find stable matching executes in $O(n)$ $O(n^2)$ time where n is the number of women.

- ☐ A S1 is false and S2 is true
- ☐ B Both S1 and S2 are true
- ☐ C Both S1 and S2 are false
- ☒ D S1 is true and S2 is false

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Question 4 In the stable marriage problem, what is the primary objective?

- ☐ A Ensure that no unstable pairs exist.
- ☐ B Ensure that everyone prefers their partner more than anyone else. ✓
- ☐ C Maximize the overall happiness. ✓
- ☐ D Minimize the number of rejections. ✓

Question 5 ♣ Which of the following statement(s) is /are true regarding the stable matching problem?

- ☒ A There won't be a pair unmatched at the end of the execution of Gale-Shapley algorithm.
- ☒ B Gale-Shapley algorithm always terminates.
- ☒ C There could be more than one stable matchings possible for a given input.
- ☒ D It is possible to have certain preference lists where the Gale-Shapley algorithm fails to give a stable matching.
- ☒ E None of these answers are correct.

Question 6 Which among the following is/are true about randomized quick sort? (Assuming n is the input size)

- ☒ A The expected running time on every input is $O(n \log n)$
- ☒ B The worst case expected running time on every input is $O(n^2)$
- ☒ C The adversary can choose an input such that the algorithm takes $O(n^2)$ time on that input.
- ☐ D On any given input, the algorithm terminates in $O(n \log n)$ time

Question 7 In a Quicksort type algorithm, if the $n/10^{\text{th}}$ largest element is chosen as the pivot in every iteration, what is the approximate height of the recursion tree?

- ☐ A $n \log n$
- ☐ B n
- ☐ C $n/\log n$
- ☐ D $\log n$

Question 8 What is the optimal time complexity to find the median of an unsorted array of size n ?

- ☐ A n^2
- ☐ B $n \log n$
- ☐ C n
- ☐ D $\log n$

Question 9 In a set of 25 identical coins, one coin is known to be counterfeit. However, its nature (heavier or lighter) is unknown. What is the minimum number of weighings needed to both identify the counterfeit coin and determine whether it is heavier or lighter than the genuine coins?

- ☐ A 4
- ☐ B 6
- ☐ C 3
- ☐ D 5

Question 10 Which among the following is asymptotically larger?

- ☐ A $O(n(\log n)^{500})$
- ☐ B $O(n \log n)$
- ☐ C $O(n^{\log_2 3})$
- ☐ D $(n^{1.00001})$

